



Bilkent University

CS 319

Deliverable-2

S1-T2-Databaseable

Hüsnü Arkca

Yunus Emre Aslan

Mehmet Dizdar

Aral Müftüoğlu

Onur Ege Yakar

1. Tech Stack and Architecture

- This project follows a three-tier architecture consisting of a frontend, backend, and database layer.
- The frontend is built with React.js and provides the user interface for registration, login, profile update, and account deletion.
- The backend is implemented using Java Spring Boot, which exposes RESTful APIs, handles authentication and CRUD operations, and connects to the database.

Architecture overview:

React (Frontend) → REST API (JSON) → Spring Boot (Backend) → JPA → PostgreSQL (Database)

2. CRUD Operations Demonstrated

The video demonstrates the complete CRUD functionality for the User Management use case.

- Create: Registering a new user account through the frontend form.
- Read: Logging in with credentials and verifying authentication through the backend.
- Update: Updating the user's password or profile information and confirming changes in the database.
- Delete: Deleting the user account and verifying that the corresponding record is removed from the PostgreSQL database.

3. Screencast Demonstration

Video Link: https://youtu.be/zTumTjHW2go?si=Z2iY0tdHTqCr_H53

The screencast shows the full workflow:

- Starting PostgreSQL, the Spring Boot backend, and the React frontend locally.
- Performing all CRUD operations through the browser interface.
- All components run locally on the machine without using any cloud services.
- React frontend runs on localhost:3000
- PostgreSQL runs on localhost:5432
- Spring Boot backend runs on localhost:8080